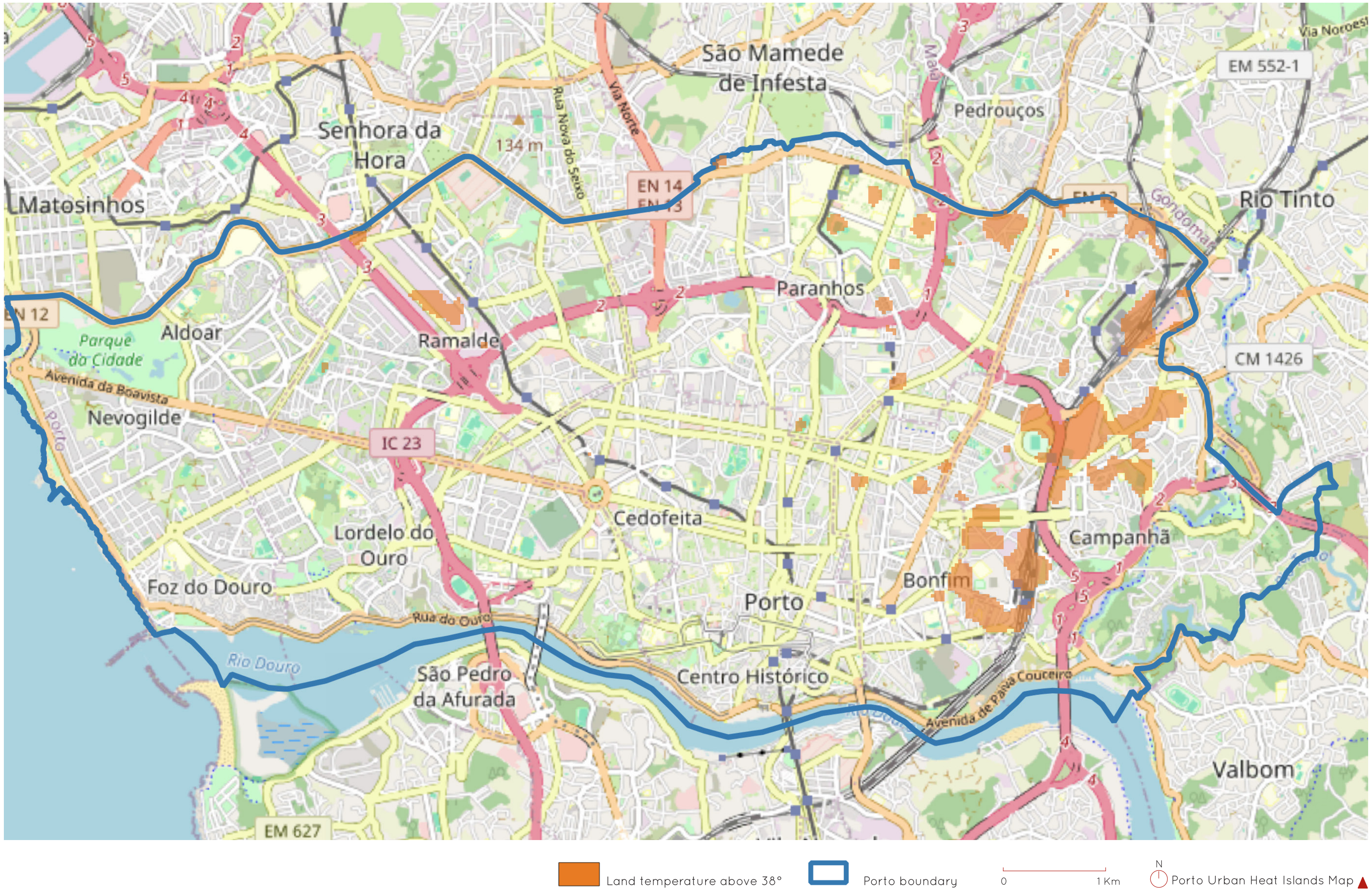
Urban Heat Island Detection in Porto

A Geospatial and Remote Sensing Analysis using Landsat Data and Python
2025, Porto

Abstract

This project maps and analyses Urban Heat Islands (UHIs) in the City of Porto using remote sensing and geospatial techniques. Landsat 8 thermal infrared imagery (Band 10) from 4 July 2024 (a hot, cloudless summer day selected to maximise the visibility of the effect) was processed in Python using rasterio, geopandas and matplotlib, and visualised in QGIS.
Land Surface Temperature (LST) was calculated by converting raw satellite radiance into brightness temperature in degrees Celsius. The raster was spatially clipped to the Porto administrative boundary, and LST values were classified into four heat zones.
The resulting thermal map reveals significant spatial disparities in surface temperature across the city, with critical zones concentrated in the East, in and around the Freguesia de Campanhã, an area with a historically more vulnerable population. The output is a classified heat stress raster that supports further spatial analysis and provides a basis for early-stage design and planning interventions sensitive to climate vulnerability.



Objectives

- Detect and visualise temperature extremes in Porto.
- Identify critical zones for urban heat.
- Classify land surface temperature into categories.
- Export results for QGIS for visualisation and future urban intervention.

Tools

- **Python:** for preprocessing and temperature mapping
- **Rasterio**: for reading and masking GeoTIFF satellite bands
- **Matplotlib:** for visualising data
- **QGIS:** map composition and geospatial visualisation
- **USGS Land Explorer:** source of Landsat thermal images
- **Landsat 8:** source of thermal infrared satellite imagery
- **GeoPandas:** for manipulating shapefiles

Methodology

Step 1: Area Selection

Acquisition of boundary for Porto filtered from national shapefile.

Step 2: Data Collection

Landsat 8 (Band 10 - thermal infrared), 4 July 2024, from USGS EarthExplorer. The date was selected to maximise the visibility of the urban heat island effect: July 2024 was an unusually warm month in northern Portugal, and 4 July specifically combined high surface temperatures with cloudless conditions over Porto, both necessary for usable Landsat thermal imagery. A hot, clear day yields the sharpest contrast between built-up zones and vegetated or water-cooled areas, which is the contrast the analysis aims to expose.

Step 3: Preprocessing

Radiance was extracted from Band10 and converted into brightness temperature.

Step 4: Classification

Temperature thresholds applied:

- Warm: 25 -30 degrees Celsius
- Very Warm: 30-35 degrees Celsius
- Hot: 35-38 degrees Celsius
- Critical: above 38 degrees Celsius

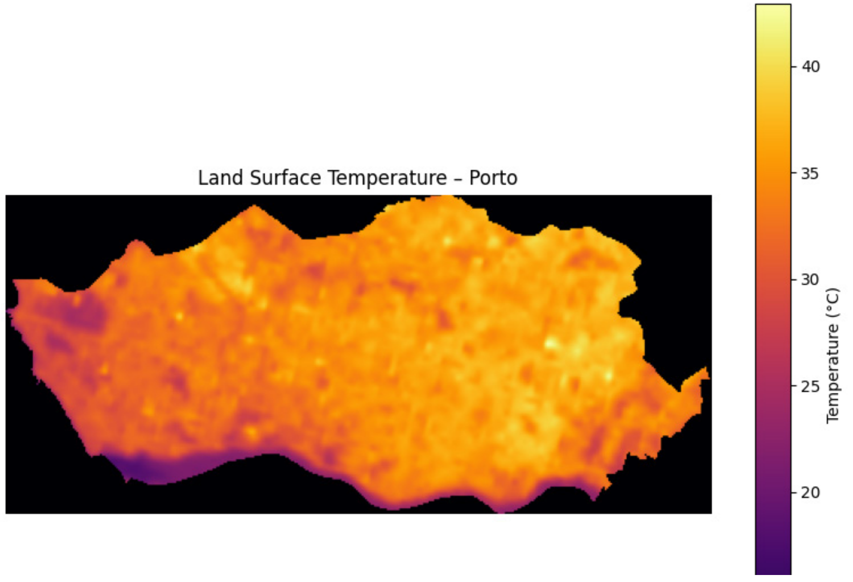
Step 5: Visualisation

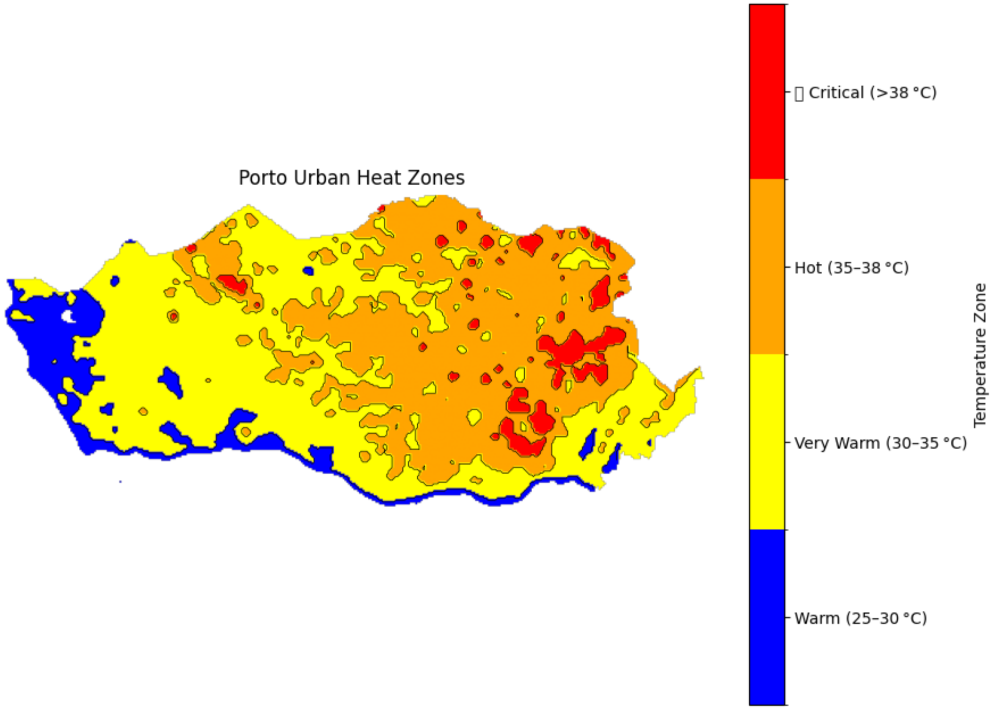
Classified raster plotted using matplotlib and QGIS.

Results

Porto's heat islands were visualised and high density critical zones identified, with special emphasis on the East of the City:

- Estação de Campanhã and surrounding areas
- Surrounding areas of Estádio do Dragão, Mercado Abastecedor and Ilhéu.
- Estação de Contumil
- Some areas in Polo Universitário S. João
- and others.





Discussion

This analysis demonstrates that publicly available data, modest computing resources, and intermediate programming knowledge are now enough to surface a problem as significant as urban heat island distribution. For architects, the implication is that this kind of geospatial reading of the territory can be brought into the early stages of project work, where design decisions still have room to respond to climate vulnerability rather than only mitigate it later.

The Porto results also carry a social dimension worth naming. The critical zones concentrate east of the city, in and around Campanhã, an area with a historically more vulnerable population. The heat is not distributed neutrally, and the populations most exposed are often those with least capacity to adapt. A planning response that does not engage with this asymmetry will reproduce it.

A single-day snapshot is, by its nature, one observation drawn from a distribution. A more robust analysis would aggregate across multiple cloudless summer days over several years, smoothing anomalies and allowing trend detection. The single-day approach was chosen for tractability, but the limitation is worth naming.

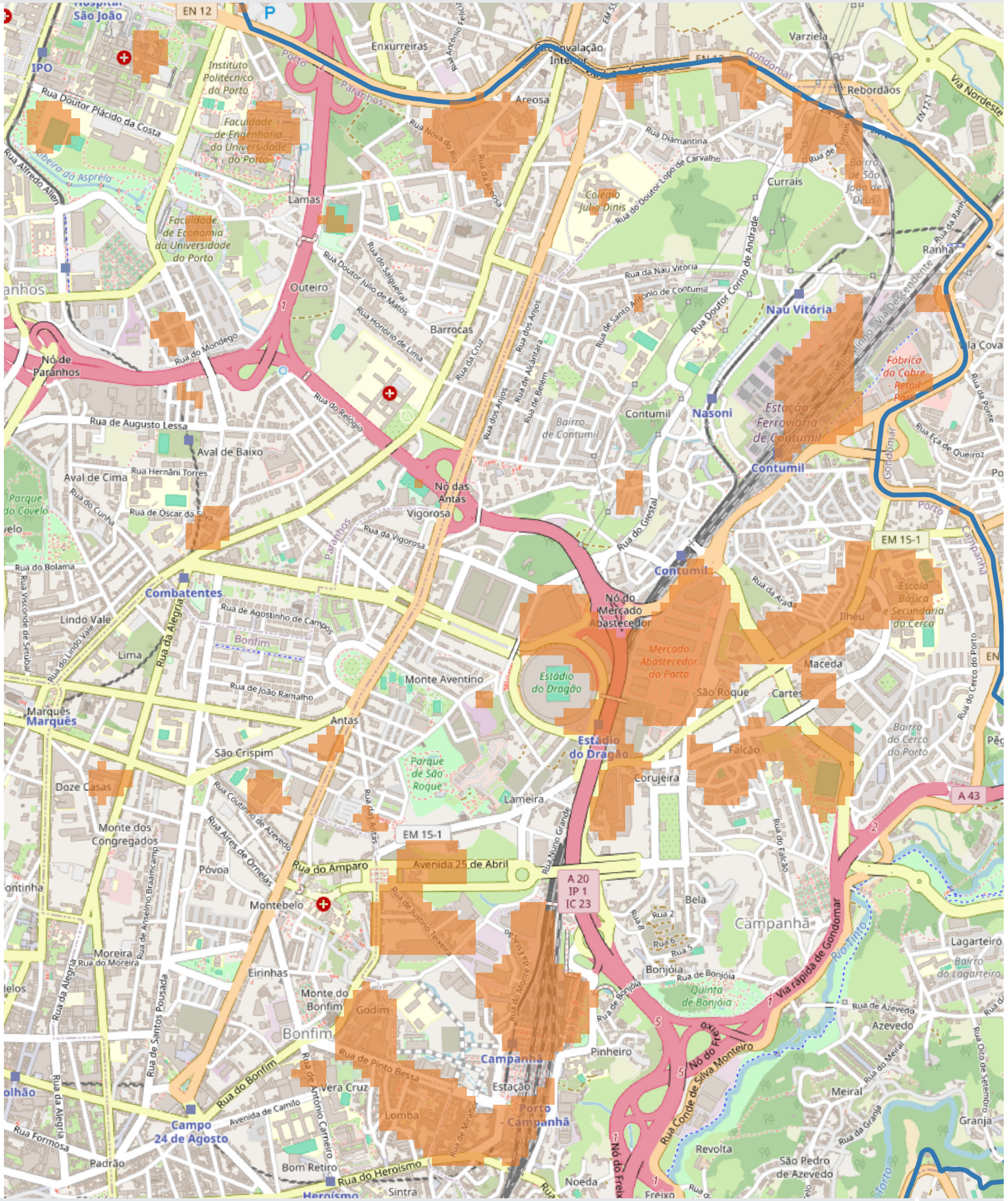
Future steps

Several directions would deepen this analysis. One would correlate the heat distribution with the Normalized Difference Vegetation Index (NDVI), quantifying the role of vegetation in mitigating surface temperature and identifying which zones would benefit most from green interventions. A second is overlaying the heat map with socioeconomic data at the freguesia or census-unit level, making the social asymmetry measurable rather than inferred. A third is temporal aggregation across multiple summer days over several years, stabilising the result and enabling trend detection, such as whether specific zones are warming faster than others. Each uses publicly available data and remains within reach of the same toolchain.

Conclusion

Urban heat islands can now be detected by the people most equipped to act on them: architects and planners. The technical barrier between climate analysis and design practice has dropped further than the practice itself has recognised. Closing that gap is partly a matter of tools, but mostly of orientation: of treating geospatial reading of the territory as part of how a project begins.

The Porto results, with their concentration in the East and the social asymmetry that follows, are a small example of what becomes visible when this orientation is adopted. They are also a reminder that the question is not only what cities should do about heat, but who decides, and on what evidence.



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Land temperature above 38° Urban Heat Islands - Porto East